

EE/CprE/SE 491 WEEKLY REPORT 2

09/29/25 – 10/7/25

Group number: 36

Project title: Secure Wireless Aircraft Data Load

Client &/Advisor: Daniel Riechers and Mohammed Selim

Team Members/Role:

Anish Nag

Gurumanie Singh

Lincoln Hatlestad

Declan Creadon

Justin Rohde

Doyle Chism

Weekly Summary

The main objective this past week was to finalize an idea of how we want to set up the testing environment. We have had a couple of meetings with our client advisor and faculty advisor to discuss various options and narrow down to the seemingly best and feasible solution. To be more specific, we discussed the feasibility of using USRPs over Raspberry Pis, estimating costs for each possible solution, and then what would be difficult and easy to implement for each solution. We eventually decided on using Raspberry Pi's and began diagramming the testing environment. We listed each component of our testing environment, both hardware and software, and the purposes, so that it is clear and specific. After discussing with the client, we have discussed with our faculty mentor and are now ready to order the hardware needed to physically set up the testing environment. As for the add-ons required for the Raspberry Pi, we are still looking into what would be the bare minimum needed to give a deliverable product to our client at the end of our project.

Past week accomplishments

- Anish Nag: Read through the codebase for implementing various encryption/hashing algorithms on the payload, checksum implementation for when the data is in transmission from point A to point B, and possible implementations of ARINC 645 concepts onto the raspberry pis. Found out areas in security where there are gaps and researched solutions to fill gaps. Main gap being the digital signature for signing the data before it is encrypted. This is the first layer of security.
- Gurumanie Singh: Read through ARINC 615A and some other documentation relating to the criteria set by Aerospace standards. Apart from this, looked into the additional things we would need to attach to our Raspberry Pi to ensure that we are able to give a proper deliverable to our client by the end of the project.
- Doyle Chism: I found and read through an EAP-TLS article process utilizing FreeRadius and DENT-Authentication to implement our authentication for the Raspberry Pi's solution. I also analyzed an open source codebase and learned more about ARINC 615A and ARINC 429, and the way we can implement the software solution when we get the Raspberry Pis.
- Lincoln Hatlestad: Researched ARINC 615A codebase. Researched different Raspberry Pi kits and RAM sizes. Created purchase orders table.
- Declan Creadon: Researched some possible raspberry pi attachments (additional memory, wifi, increased range, etc.). Tried to figure out the size of the data we are sending to get an idea of the constraints we will be under. Analyzed some open source codebases related to ARINC 615A.
- Justin Rohde: Studied the specifics of ARINC 655 regarding the requirements of the packages to be loaded. Researched potential ad ons for the raspberry pi. I specifically researched raspberry pi antenna kits and SSD drives for optimized performance and better storage. I also read more into EAP-TLS and other implementations of it.

Pending issues

- 1(Anish Nag): Found a gap in security that needs to be figured out: digital signing of the payload. ARINC 645 does not have this aspect implemented. Over the next week, need to propose a solution that fills this gap. Specifically a solution that does not go

overboard. Efficiency over complexity.

- 2 (Gurumanie Singh): Getting a green light from the client with regards to the implementation and if it lines up with what he thinks the solution should look like has been one of the main concerns. Since the client was unable to make it to the meeting last week and has not responded to any emails so far, we are still waiting on his input on some specific issues at hand.
- 3(Doyle Chism): After researching ARINC 429, I am unsure if it is necessary for our implementation. I am waiting for approval from the client to see if we should proceed with this security process.
- 4(Declan Creadon): Need to figure out how we are going to secure the network our pi's will transmit on. VPN could be possible.
- 5(Justin Rohde): Need to order raspberry pis and confirm with the client to proceed with our proposed implementation.

Individual contributions

<u>NAME</u>	<u>Individual Contributions</u> (Quick list of contributions. This should be short.)	<u>Hours this week</u>	<u>HOURS cumulative</u>
Anish Nag	● Read codebase for ARINC 645 ● Decide whether TFTP is a viable option instead of switching to SFTP ● Researching IEEE Wifi6 and WPA3 security and how it is possible to implement onto raspberry pis ● Diagramming how sha256 and crc16 will be used when the payload is in transmission ● Proposing solution for digitally signing payload before encrypting/hashing	6	22
Gurumanie Singh	● Client communication ● Research on ARINC 615A ● Add ons for Raspberry Pi ● Our implementation strategy	5	15
Doyle Chism	● Researched EAP-TLS implementation ● Researched on ARINC 615A and 429 ● Researched Raspberry Pi's add-ons for future implementation	4	14

Lincoln Hatlestad	● Researched ARINC 615A codebase ● Researched different Raspberry Pi kits and RAM sizes ● Created purchase orders table	3	15
Declan Creadon	● Researched Pi attachments ● Researched possibility of VPN ● Looked at some codebases	3	14
Justin Rohde	● Studied the specifics of ARINC 655 regarding the requirements for the packages to be loaded. ● Researched potential add-ons for Raspberry Pis. ● Specifically looked into Raspberry Pi antenna kits for improved wireless performance. ● Investigated SSD drives to optimize Raspberry Pi performance and storage capacity. ● Read more about EAP-TLS and other implementations for secure wireless communication.	5	18

Comments and extended discussion

Our meeting with the client advisor was cancelled. Therefore we were unable to confirm with our client advisor if we have the greenlight to order hardware. This pushes back our goals by one week. Therefore we need to get confirmation as soon as possible that the most up to date proposal for hardware is ok before we order.

Plans for the upcoming week

- Anish Nag : Start taking chunks and pieces from ARINC 645 codebase and putting them together to work with raspberry pi implementation. The specific chunks will be the sha256 hash generator and CRC 16 checksum generators. Generate test payload, preferably small data in the beginning to see if it works at the bare minimum level. Scale our way up to test limits. This is the main task for me because it is the foundation therefore I want to focus my time on making the foundation strong.
- Gurumanie Singh: Ensure that we confirm our implementation of how we want the solution to look like, get the green light from the client, confirm our add ons for Raspberry Pi, and ensure that we place the order by end of week.
- Doyle Chism: Continue to analyze the codebase that our client provided, so once

we know which implementation we are choosing, I can be prepared for the next steps. As a team, we need to all agree on what exact items we need to purchase, so once the client approves them, we can purchase them and begin working with our Raspberry Pi.

- Lincoln Hatlestad: Attempt to fix CMake errors on laptop to successfully run the open source ARINC 615A Data Loader and Data Loader Target. Isolate what components of the software are necessary, unnecessary, and need to be built for our solution. Research possible hardware add-ons to present to our client.
- Declan Creadon: Want to figure out how we will make the Pis communicate. Will it be over wifi, will it be cellular, satellite, etc. I also need to figure out how to tune frequencies and other components to get a reliable connection.
- Justin Rohde: Order raspberry pi. Load up raspberry pi virtual machine on my local machine to start working on potential code implementations. Start working on EAP-TLS infrastructure on the VMs. Establish initial connection across multiple raspberry pis through virtual machines.

Summary of weekly advisor meeting

In our meeting with Professor Selim this week, we reviewed what we did since our last meeting, asked some questions about budget, and discussed possible questions for the client. Below is a more specific summary of the meeting.

Guru explained how we split the team's resources on researching USRPs vs. Raspberry Pis, and his own research on EAP-TLS and the ARINC 429 codebase. Anish explained his findings on the ARINC 645 protocol and codebase, also Trivial File Transfer Protocol, and explained the technical details to the group. Justin explained that he researched the ARINC 665 protocol and codebase. During Justin's review, the team brought up concerns about lack of specific requirements from the client. Selim suggested we research the benefits of different Pi add-ons while we wait for requirements. Declan explained that he researched ARINC 665 AND 645 with Anish and Justin. During Declan's review, the type of wireless solution was brought into question. We ultimately decided on network based. We ran out of time before Doyle and Lincoln got a chance to review their work.